



Genital Manifestations of Physical Abuse in Male Infants and Children

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We present 3 male patients with genital bruising due to physical abuse to improve recognition of genital trauma as a sentinel injury. In the absence of an underlying medical condition or a clear acceptable accidental mechanism for the genital injury, an evaluation for child abuse is recommended. UROLOGY 178: 151–154, 2023. © 2023 Elsevier Inc. All rights reserved.

The infrequency of bruising in non-mobile infants was highlighted in the seminal paper by Sugar et al. entitled “Bruises in Infants and Toddlers: Those Who Don’t Cruise Rarely Bruise.” Out of a cohort of 366 children less than 6 months of age presenting for well-child care, just 2 children (0.6%) had bruising.¹ Given its rarity in pre-cruising infants, further research has underscored the high prevalence of bruising in young children who have been physically abused. Studies examining the location and features of bruising in children with suspected abuse versus accidental injury have identified characteristics that can help distinguish between the two. Most recently, Pierce et al. published a study validating a bruising clinical decision rule (BCDR) for the recognition of physical abuse (PA). The BCDR, known as TEN-4-FACESp, refers to any bruising in an infant 4.99 months or younger and bruising of the torso (including buttocks, genitals, anus), ear, neck, frenulum, angle of jaw, cheeks, eyelids, subconjunctival hemorrhage, and patterned bruising in a child younger than 4 years. Bruises in these regions indicate a potential risk of abuse with bruising to the buttocks, genitourinary area or anus, angle of the jaw, neck, and subconjunctival hemorrhage being the most specific regions for abuse.²

In this case series, we describe 3 male patients with inflicted genital injuries, each of whom represented to medical care across various settings with additional, severe injuries because of either missed or delayed recognition of the presenting genital injury as suspicious for abuse. Case 3 has been presented as a figure in the textbook Diagnostic Imaging of Child Abuse.³

CASES

1. A 4-week-old former full-term male was referred to pediatric urology clinic for evaluation of penile discoloration

and bruising. The day prior, the infant had been seen at an emergency department (ED) after his parents noted redness and swelling of his penis and a small amount of bright red blood in the diaper. A history of trauma was denied but the parents wondered whether they could have clipped his penis when buckling him into his car seat the previous day. In the ED, he had a small erosion on the dorsal surface of the glans with swelling of the shaft of the penis, a cluster of erythematous macules on the dorsum of the shaft, and mild erythema of the tip of the penis. A presumptive diagnosis of lysis of adhesions was made. The following morning the infant was evaluated by his primary care provider (PCP) where petechiae were visualized along the shaft of his penis with “dark” erythema of the skin as well as erythema of the glans and he was booked for a same-day appointment with pediatric urology. Aside from a resolving hematoma and bruising of the penis, the exam in the urology clinic was otherwise unremarkable, and the family was discharged.

One month later, the child returned to the same ED with abusive head trauma (AHT). Upon transfer to a pediatric hospital, further evaluation guided by the Child Protection Team (CPT) revealed bilateral retinal hemorrhages, 12 fractures, a spinal subdural hemorrhage, and scarring of his lingual frenula. The Department of Children and Families (DCF) assumed custody of the patient and he was discharged to a foster placement.

2. A 2-month-old former full-term male was transferred from an outside hospital with bilateral mixed-density subdural hemorrhages and altered mental status. His father reported that the patient had fallen from a bed onto carpeted flooring. On exam in the pediatric ED, the infant had “slight purplish discoloration” to his right scrotum. The infant’s mother recalled 2 prior episodes of bruising to his right scrotum within the preceding 1 month. On the first occasion, mother did not seek medical care. When the bruise “reappeared,” mother brought the infant to be evaluated by his PCP, as he was also vomiting. Review of the outside records was notable for a normal scrotal and pyloric ultrasound (US). Upon admission, the infant underwent additional evaluation by CPT. The history of a short fall onto carpeted flooring was felt to be incompatible with the severity of the

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Table 1. Compilation of cases both previously published and currently of genital manifestations of physical abuse in male infants and young children.

Age	Evaluation Location	Physical Exam Findings	Documented Delayed Recognition/Missed Abuse	Consequences of Delayed Diagnosis of Abuse
3 week old ¹²	ED	Acute swelling and bluish discoloration of the left hemiscrotum; left inguinal fullness; distended abdomen	No	-
4 week old	PCP, urology clinic, and ED	First presentation (multiple different providers over a period of days): erythema, swelling, and discoloration of the shaft and glans penis consistent with bruising Second presentation (1 month later): lethargy Altered mental status; slight purplish discoloration to the right scrotum	Yes	Child represented with AHT and numerous skeletal injuries
2 month old	ED	First presentation: Superficial irritation and abrasions of the knees, lower legs, penis, and scrotum from being left for prolonged periods on a rough crib sheet Second presentation (1 month later): failure to thrive with a 0.8 kg weight loss	Yes	Child presented with AHT
2 month old ¹⁰	ED and inpatient setting	Small purpuric lesion beneath the left eye; erythema of the glans penis with abrasions and contusions from the orifice to the ventral edge 75% circumferential laceration to the base of the penis	No	-
4 month old ¹⁰	ED	First presentation: large, unexplained subgaleal hematoma Second presentation (10 days later): purpuric lesions to the shaft and glans of the penis	No	-
11 month old ¹⁰	Not described	Third presentation (4 days later): lethargy, bilateral eyelid swelling; hematomas to the left buccal mucosa, left forehead, left upper eyelid, and knees; dark scalp discoloration; bilateral corneal abrasions; edematous/erythematous lips; penile swelling with healing and acute hematomas to the glans and shaft	No	-
14 month old ¹⁰	ED and inpatient setting	Multiple presentations for scrotal erythema and swelling	Yes	Child represented with more severe cutaneous and ocular injury
19 month old ³	ED, PCP, inpatient pediatric floor, urology clinic	Suprapubic bruising and penile swelling (Fig. 1)	Yes	Child represented on multiple occasions with recurrent injury culminating in severe immersion burns to his lower extremities
2 year old	ED	Contusions of the left forehead, supraorbital region, and upper lip; swelling, engorgement, and abrasions to the distal penis that was initially attributed to the child falling but later to being "chastised" by his father	No	-
2 year old ¹⁰	ED		No	-

Table 1 (Continued)

Age	Evaluation Location	Physical Exam Findings	Documented Delayed Recognition/Missed Abuse	Consequences of Delayed Diagnosis of Abuse
3 year old ¹⁰	ED	Swollen and erythematous scrotum after being spanked with a belt by his father	No	-
4 year old ¹¹	Not described	Facial swelling and respiratory distress; scrotal ecchymosis; missing mandibular incisors	No	-

The table highlights missed opportunities for intervention and the resultant consequences.

child's intracranial injuries and a diagnosis of AHT was made. A mandated report for PA was filed and the infant was removed from the parents' custody.

3. A 19-month-old male arrived at an outside ED with bilateral scrotal erythema and swelling for which he was presumptively diagnosed with infection following an US and discharged with a prescription for antibiotics. The redness continued to spread despite treatment, and he presented to a tertiary children's hospital with swelling and ecchymosis over the entire scrotum. Pediatric urology was consulted, a repeat US was performed, and Henoch Schonlein purpura (HSP) was suggested as a possible diagnosis. The child was discharged with plans for follow-up with his PCP and pediatric urology and was instructed to continue the antibiotics. However, the child's symptoms recurred and over the course of a month, he had 3 admissions to the hospital. The third time, he returned with pronounced swelling of his right hemiscrotum prompting surgical exploration for a presumed scrotal abscess. Instead, surgery revealed a hematoma in the right hemiscrotum. After CPT consultation, a skeletal survey identified a healing fracture of the right distal radius and ulna, a healing supracondylar fracture of the left distal humerus, as well as a swallowed foreign body resembling a battery. A filing for neglect, rather than PA, was made on behalf of the child given his ambulatory status and DCF-sanctioned discharge to the mother's care.

Two months later, the child returned to the ED with clearly demarcated, bilateral lower extremity burns consistent with forced immersion as well as bruising to his forehead and a linear parietal skull fracture.

DISCUSSION

Genital bruising in children is rare, even in those with an underlying bleeding disorder.⁴

The differential diagnosis for genital bruising in young children includes intra-abdominal hemorrhage, HSP, and direct trauma to the area.⁵⁻⁸ Traumatic causes of bruising to the male genitalia may be accidental or non-accidental. The literature on the latter is limited or views genital bruising only through the lens of sexual abuse rather than PA.⁹ A case series from 1978 described 6 boys (2 months to 3 years of age) with penile trauma, 5 with bruising, and 1 with laceration.¹⁰ All were found to have other signs of abuse. The authors concluded that traumatic penile lesions should raise concern for inflicted injury. Table 1 presents a summary of the case series from 1978, other single case reports in the literature,¹⁰⁻¹² and the cases presented in our series, as well as a companion case in which abuse was recognized at the time of initial presentation.

A complete genital exam in a child with genital bruising includes inspection of the meatus, the frenulum which lies immediately below it, the glans, the penile shaft skin, the scrotum, and the scrotal contents. For



Figure 1. A 2-year-old male with delayed speech presented to the ED for evaluation of suprapubic bruising and penile swelling. The mother denied any known antecedent trauma but wondered whether it occurred when the child's 11-year-old sibling unbuckled the patient from his car seat the day prior. On external genital exam, there was a 3 × 2 cm area of suprapubic tissue with blotchy, blue, and purple ecchymosis and surrounding erythema. Additionally, a semi-circumferential linear mark was present on the right of the penile shaft measuring 1 cm in length. The dorsum of the penile shaft also had blue ecchymosis. A mandated report was filed for PA due to the extensive genital bruising and lack of a plausible explanation in a non-verbal child. (Color version available online)

uncircumcised infants, the glans may not be fully visible, and attempts to retract the foreskin should be made only gently. Penile bruising should be distinguished from penile edema alone, as the latter has a myriad of causes. A head-to-toe assessment should also be performed, to not overlook concurrent traumatic injury.

A comprehensive history should be obtained from the patient (when able) to determine whether the proposed mechanism may be plausible. Children presenting with genital bruising, in the absence of an underlying medical condition or a clear, acceptable accidental mechanism, should undergo further evaluation for child abuse.¹³ A multidisciplinary evaluation in consultation with a child abuse pediatrician is recommended.

Declaration of Competing Interest

The authors have no conflicts of interest to disclose.

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